

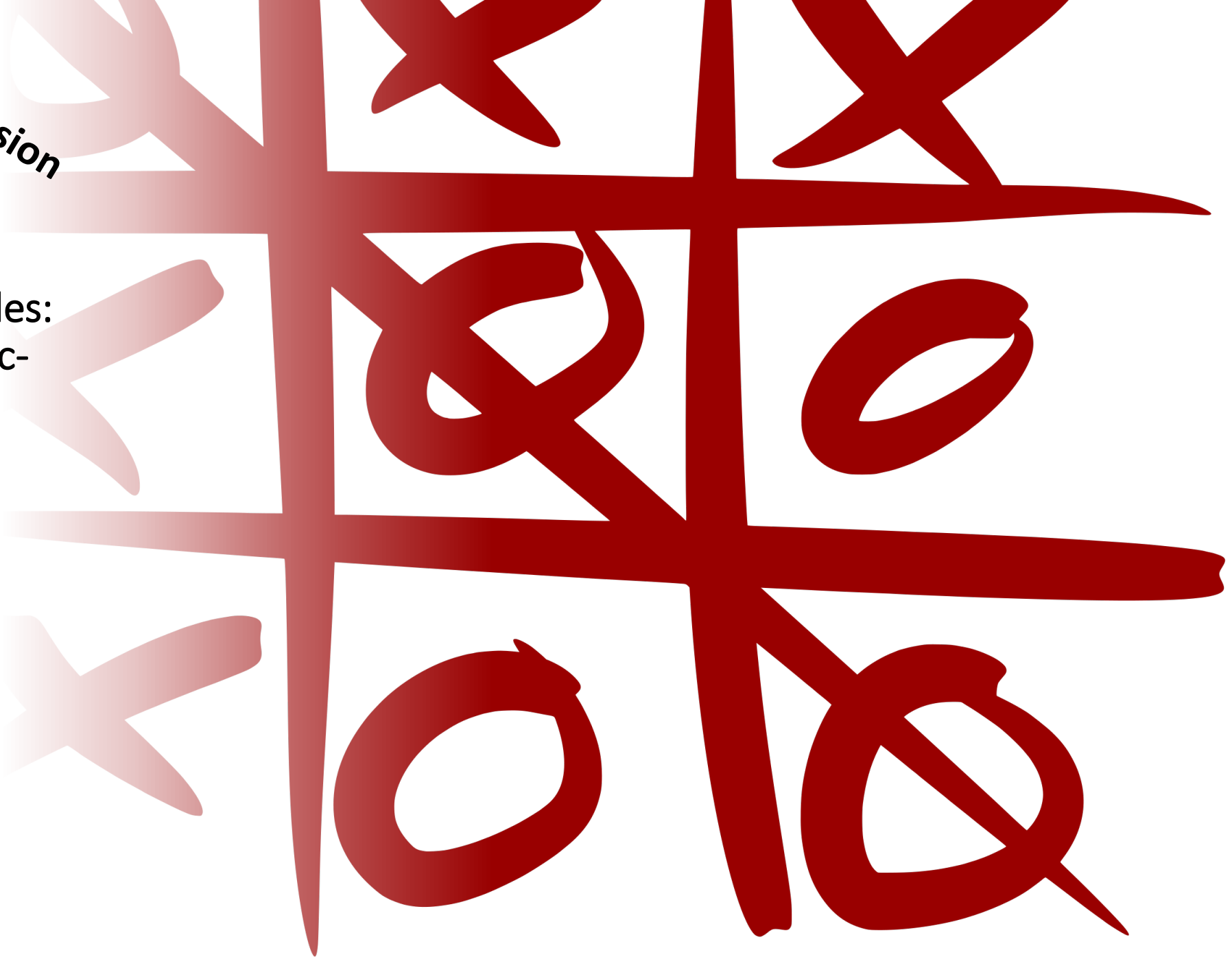
Discussion

CS 5/7320
Artificial Intelligence

Learning from Examples:
Learning to Score a Tic-
Tac-Toe board

AIMA Chapter 19

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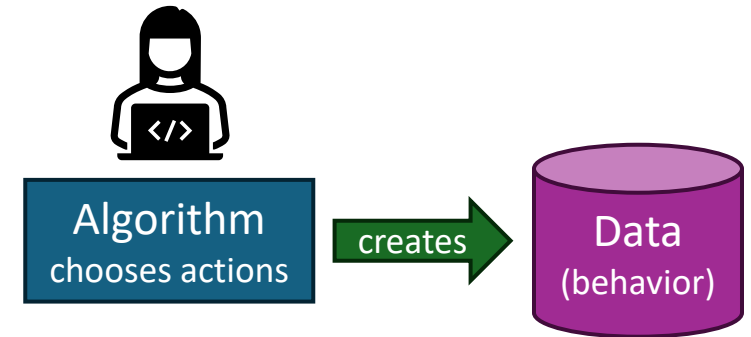
Intro: Learning from Examples

What can be learned?

Learning from Examples

Up until now in this course:

- **Hand-craft algorithms** to make rational/optimal or at least good decisions.
Examples: Search strategies, heuristics, and constructing Bayesian networks.

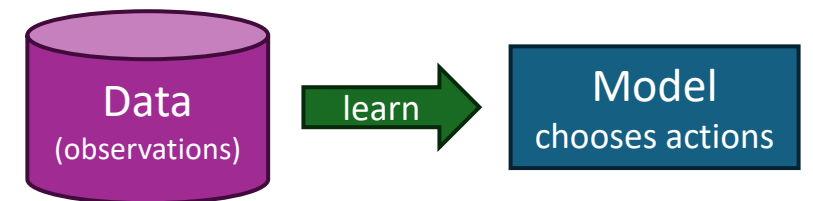


Issues

- We may not be able to anticipate all possible future scenarios.
- We may have examples, but we do not know how to implement a solution.

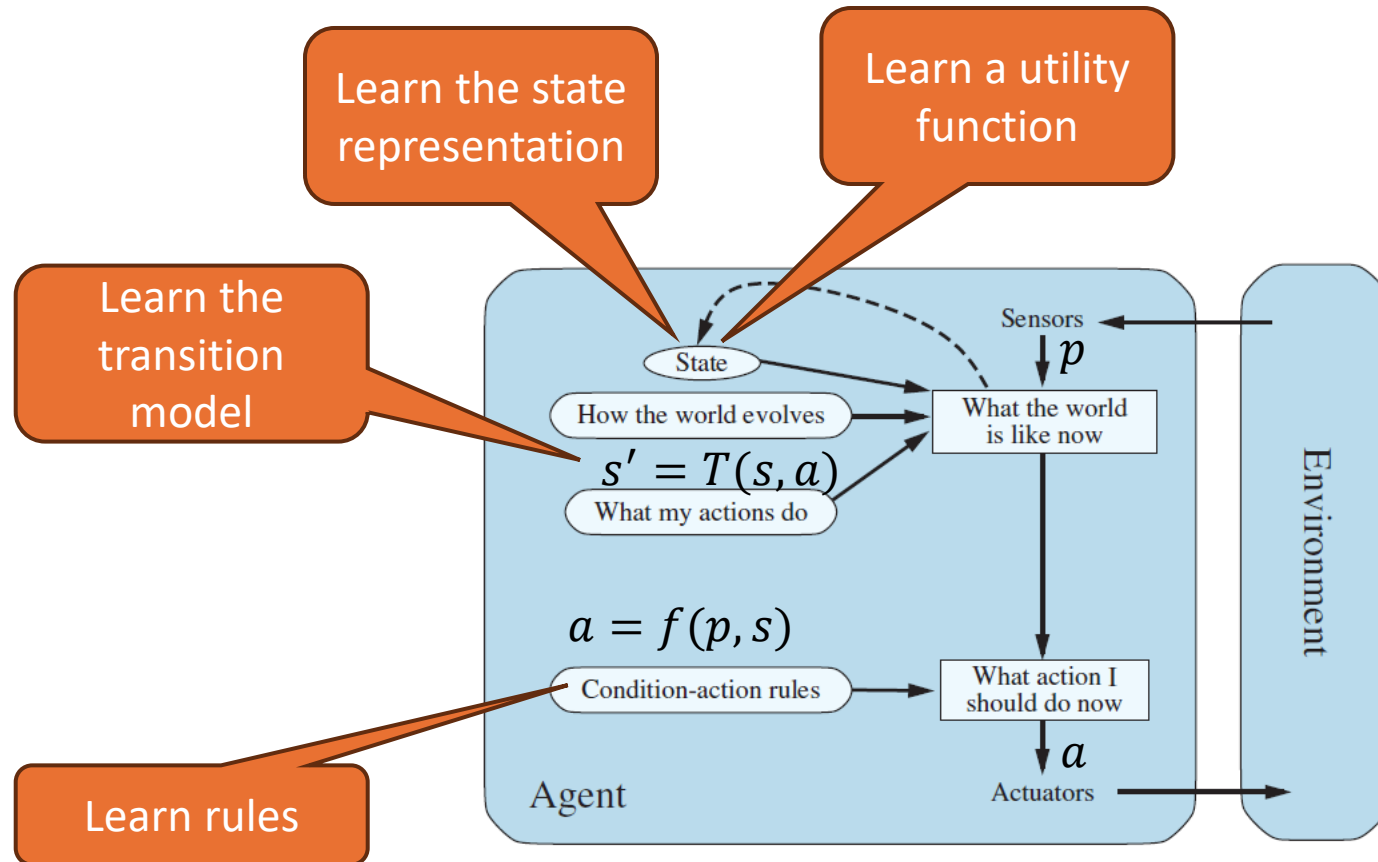
Supervised Machine Learning

- Uses observations: training data with the correct answers.
- Learn a function (model) to map an input (e.g., state) to an output (e.g., action) representing the desired behavior.
- Examples:
 - Use a naïve Bayesian classifier to distinguish between spam/non-spam.
 - Learn a playout policy to simulate games (current board -> good move)



Learning Components of an Agent

- We can learn many different components of an agent from examples
- **Example:** Learning components of a model-based reflex agent



Typical Use of Supervised ML for Intelligent Agents

Learn a Policy

- Classification: Directly learn the best action for each state from examples.

$$a = h(\text{state features})$$

- This model can also be used as a **playout policy** for Monte Carlo tree search with data from self-play.

Learn Evaluation Functions

- Regression: Learn evaluation functions to estimate state utilities.

$$\text{eval} = h(\text{state features})$$

- Can learn a **heuristic** for heuristic alpha-beta search.
- For reinforcement learning we can learn action values $q(\text{state}, \text{action})$.

Learn Perception and Actuation

- **Natural language processing:** Use deep learning / word embeddings / language models to understand concepts, translate between languages, or generate text.
- **Speech recognition:** Identify the most likely sequence of words.
- **Vision:** Object recognition in images/videos. Generate images/video.
- **Robotics:** Learn how to move safely.

Compressing Tables

- Neural networks can be used as a compact representation of tables that do not fit in memory. E.g.,
 - Joint and conditional probability tables
 - State utility tables (i.e., an evaluation function)
 - Q-Value tables in reinforcement learning

Bottom line: Learning a function is often more effective than hard-coding it. However, we do not always know how it performs for rare and edge cases!

Example: Tic-Tac-Toe

Approaches to implement/learn an evaluation function.

Utility of Being in a State: State Value

State s



$U(s)$

Approach: Try all actions and pick the action that leads to the state with the largest state value.

$$\operatorname{argmax}_{a \in \mathcal{A}(s)} U(\operatorname{result}(s, a))$$

The state value is defined as:

$$U(s) = \mathbb{E}_{\pi_x, \pi_o}[u | s_0 = s],$$

with the playout policies π_x, π_o of player x and o .

Questions:

- What playout policies do we use? Random, optimal, ...
- How do we estimate the expected value?

u is the observed utility of when finishing a game starting at board s .

Expectation is used for stochastic playout strategies!

Minimax and Variations

$$U(s) = \mathbb{E}_{\pi_x, \pi_o}[u | s_0 = s]$$

Minimax replaces
the expectation with
optimal play!

State s	$\text{minimax}(s)$			
	1			
	0			
$\vdots$				

Playout policy

π_x, π_o

Method

optimal play

Tree search

Expert Evaluation Function

$$U(s) = \mathbb{E}_{\pi_x, \pi_o} [u | s_0 = s] \approx \textit{eval}(s)$$

State s	$\textit{minimax}(s)$	Expert code $\textit{eval}(s)$		
	1	+0.4		
	0	-0.4		
⋮				

Expert Heuristic

+0.4 for each winning opportunity
-0.4 for each losing opportunity

Playout policy

π_x, π_o

Method

optimal play

none

Tree search

Code + cutoff
search

Monte Carlo Simulation

$$U(s) = \mathbb{E}_{\pi_x, \pi_o}[u | s_0 = s]$$

Monte Carlo Sim.

$$\approx \frac{1}{N} \sum_{i=1}^N u_i | s_0 = s$$

State s	$\text{minimax}(s)$	Expert code $\text{eval}(s)$	Monte Carlo simulation	
	1	+0.4	0.634	
	0	-0.4	-0.021	
$\vdots$				

Playout policy
 π_x, π_o

optimal play

none

Random or
other

Method

Tree search

Code + cutoff
search

Simulation

Machine Learning

$$U(s) = \mathbb{E}_{\pi_x, \pi_o} [u | s_0 = s] \approx \mathbf{h(s)}$$

State s	$\text{minimax}(s)$	Expert code $\text{eval}(s)$	Monte Carlo simulation	Machine Learning $h(s)$
	1	+0.4	0.634	0.325
	0	-0.4	-0.021	0.042
⋮				

Playout policy
 π_x, π_o

optimal play

none

Random or
other

Used for simulating
or training data

Method

Tree search

Code + cutoff
search

Simulation

Function
approximation

All methods
hopefully produce a
consistent state
order.

Reinforcement Learning (RL)

Tries to learn state values directly from rewards resulting from

- **Model-based RL:** Uses a complete probabilistic model of the environment.
- **Model-free RL:** Learns from interactions with the environment.
 - Online learning
 - Offline learning (can use supervised learning from a sampled dataset)

Machine learning for an evaluation function can be seen as a special case of model-free offline RL called (neural) fitted Q-learning with replay memory.

You can learn more by taking an RL course.